IEEE Journal of Emerging and Selected Topics in Industrial Electronics

Special Section on:

Solar Photovoltaic Integration to Power Grid: Potential Issues and Uprising Technologies

Solar photovoltaic integration into power systems is rapidly emerging as a transformative solution for the global energy transition to support the vision of net zero and mission of decarbonization. However, issues exist as Solar PV is only available during the daytime and reduces system inertia. The weather dependency and intermittency call for expensive storage accessories and use of accurate forecasting methodologies. Impact of energy transactions and energy market of such renewable energy integration needs extensive investigation. Solar PV in connection with electric transportation is another area where a development of solar PV fed microgrid with proper energy storage may promote EV charging in remote locations. Efforts are ongoing to develop converter systems that are reconfigurable to enhance performance and stability. Grid Forming Control (GFM), Grid Following Control (GFL), their hybrid, Virtual Synchronous Generators (VSG) etc. are proposed to handle such systems. This special section will explore upcoming technologies and potential solutions to all the issues for massive Solar PV integration to the power grid. Key focus areas of this special section will include power electronics, optimization, forecasting techniques, computational intelligence, hybrid energy networks, advanced control strategies for enhanced stability, and resiliency, evolving power grid to avoid congestion, and energy storage.

We encourage all researchers working in this area to submit papers to this Special Section. Topics of interest include, but are not limited to:

- Advanced power electronics for efficient PV-grid interaction
- Improving reliability and mitigating degradation in PV systems
- Testing and standardization of PV modules for long-term reliability
- GFM, GFL and Hybrid control for improving stability
- Reconfigurable converter topologies and controllers for boosting system inertia
- Forecasting methods and energy markets for promoting Solar PV systems
- Solar PV for Grid to Vehicle and Vehicle to Grid applications
- Use of AI and machine learning to manage PV intermittency
- Smart PV technologies with embedded sensors and AI-driven monitoring
- Advanced sensing, monitoring, and maintenance of solar panels
- Modeling, digital twins and real-time simulation for large-scale hybrid energy networks

Manuscript Preparation and Submission

Check carefully the style of the journal described in the guidelines "Information for Authors" in the IEEE - IES website: http://www.ieee-ies.org/pubs/jestie. Please submit your manuscript in electronic form through: https://mc.manuscriptcentral.com/jestie-ieee/.

On the submitting page, in pop-up menu of manuscript type, select: "**SS on Solar Photovoltaic Integration to Power Grid: Potential Issues and Uprising Technologies**", then upload all your manuscript files following the instructions.

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Timetable

Deadline for manuscript submissions:	Information about manuscript acceptance:	Publication Date:
May 31, 2026	**October, 2026**	**Jan, 2027**

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